Geon-Woo Kim

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Research Interests

Systems for Machine Learning, LLM Training and Inference at Scale

Education

The University of Texas at Austin — Austin, TX, USA
PH.D. Student in Computer Science — *Sep 2022 - Present*
Advisor: *Aditya Akella and Daehyeok Kim*

Seoul National University — Seoul, South Korea
B.S. in Computer Science & Engineering — *Mar 2013 - Aug 2022*
B.S. in Mathematical Sciences (Double Major)

- Summa Cum Laude. This period overlapped with five years of industrial work and alternative military service.

Publications

Reforge: Low-Latency Distributed GNN Serving with Selective Embedding Recomputation, *Geon-Woo Kim*, Donghyun Kim, Jeongyoon Moon, Henry Liu, Tarannum Khan, Anand Iyer, Daehyeok Kim, Aditya Akella, IEEE International Parallel and Distributed Processing Symposium (**IPDPS 26**)

HALoS: Hierarchical Asynchronous Local SGD over Slow Networks for Geo-Distributed Large Language Model Training, *Geon-Woo Kim*, Junbo Li, Shashidhar Gandham, Omar Baldonado, Adithya Gangidi, Pavan Balaji, Zhangyang Wang, Aditya Akella, Forty-second International Conference on Machine Learning (**ICML 25**)

StitchLLM: Serving LLMs, One Block at a Time, Bodun Hu, Shuozhe Li, Saurabh Agarwal, Myungjin Lee, Akshay Jajoo, Jiamin Li, Le Xu, *Geon-Woo Kim*, Donghyun Kim, Hong Xu, Amy Zhang, Aditya Akella, The 63rd Annual Meeting of the Association for Computational Linguistics (**ACL 25**)

Read-ME: Refactorizing LLMs as Router-Decoupled Mixture of Experts with System Co-Design, Ruisi Cai, Yeonju Ro, *Geon-Woo Kim*, Peihao Wang, Babak Ehteshami Bejnordi, Aditya Akella, and Zhangyang Wang, Thirty-Eighth Annual Conference on Neural Information Processing Systems (**NeurIPS 24**)

Lovelock: Towards Smart NIC-hosted Clusters, Seo Jin Park, Ramesh Govindan, Kai Shen, David Culler, Fatma Özcan, *Geon-Woo Kim*, and Hank Levy, HotCarbon Workshop on Sustainable Computer Systems (**HotCarbon 24**)

Orca: A distributed serving system for Transformer-Based generative models, Gyeong-In Yu, Joo Seong Jeong, *Geon-Woo Kim*, Soojeong Kim, and Byung-Gon Chun, USENIX Symposium on Operating Systems Design and Implementation (**OSDI 22**)

Terra: Imperative-Symbolic Co-Execution of Imperative Deep Learning Programs, Taebum Kim, Eunji Jeong, *Geon-Woo Kim*, Yunmo Koo, Sehoon Kim, Gyeong-In Yu, and Byung-Gon Chun, Thirty-Fifth Annual Conference on Neural Information Processing Systems (**NeurIPS 21**)

Apache Nemo: A Framework for Building Distributed Dataflow Optimization Policies, Youngseok Yang, Jeongyoon Eo, *Geon-Woo Kim*, Joo Yeon Kim, Sanha Lee, Jangho Seo, Won Wook Song, and Byung-Gon Chun, USENIX Annual Technical Conference (**ATC 19**)

Pado: A data processing engine for harnessing transient resources in datacenters, Youngseok Yang, *Geon-Woo Kim*, Won Wook Song, Yunseong Lee, Andrew Chung, Zhengping Qian, Brian Cho, and Byung-Gon Chun, ACM European Conference on Computer Systems (**EuroSys 17**)

Research Experience

Graduate Research Assistant, UTNS — UT Austin
Advisor: *Aditya Akella* — *Sep 2022 - Present*

- *HALoS*: Addresses challenges of geo-distributed and heterogeneous resource problem in LLM training through hierarchical and asynchronous training framework.
- *Reforge*: Achieved 10×-100× lower GNN serving latency with dependency-aware approximation and communication-efficient distributed execution.

Student Researcher, SystemsResearch@Google — Google
Mentor: *Seo Jin Park* — *May 2023 - Jul 2023*

- *Efficient MoE-LLM Serving*: Demonstrated, through simulation, the optimized throughput in MoE-based LLM serving using fine-grained expert disaggregation.

Undergraduate Research Intern, SPL — SNU
Advisor: *Byung-Gon Chun* — *Jan 2015 - Aug 2021*

- *Orca*: Proposed ***continuous batching***, a standard batching technique in LLM serving. Contributed to refactoring attention CUDA kernels to allow advanced KV cache management. Developed distributed deployment and fast C++-based web frontend.
- *Terra*: Enabled symbolic execution for imperative deep learning programs. Contributed to improving formal guarantees with JIT compilation and mathematical proof. Extensively profiled and optimized the system.

Work Experience

Software Engineer Intern, Machine Learning (PhD) — Meta Platforms, Inc.
Mentor: *Saif Hasan* (AI Networking Team) — *May 2025 - Aug 2025*

- Developed **fully functional NVIDIA–AMD heterogeneous GPU collectives** (AllReduce and AllToAll)
- Made Meta's internal collective framework 100% AMD-compilable
- Developed an RDMA performance tool for hetero vendor environments (e.g., NVIDIA–AMD, Mellanox–Broadcom)

Software Platform Engineer — Viva Republica (Toss)
South Korea's leading fintech startup valued at over $7 billion with over 20 million users — *Jun 2016 - Feb 2021*

- Initiated, designed, implemented, and maintained a comprehensive mobile marketing platform with multiple functionalities, including running A/B tests, sending large numbers of messages to target groups, and dynamically prompting dialogues based on real-time user logs.
- Built and maintained a large-scale data analysis platform utilizing various open-source projects, including Apache Spark, HDFS, Apache Impala, Apache Flink, and PyTorch.

Honors & Awards

Aug 2022	**Overseas PhD Scholarship**	Korea Foundation for Advanced Studies
Mar 2013	**Presidential Science Scholarship**	Korean Government

Technical Skills

Languages: Python, C++, C, Kotlin, Java

Frameworks: PyTorch, Tensorflow, DeepSpeed, DeepSpeed-MII, vLLM, Deep Graph Library, Spark, Flink